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FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

2025
TEST 1

INSTRUCTIONAL PROGRAMME :	BACHELOR OF ICT IN APPLICATIONS DEVELOPMENT/ DIPLOMA IN ICT IN APPLICATIONS DEVELOPMENT (4-YEAR EXTENDED PROGRAMME)
INSTRUCTIONAL OFFERING :	APPLICATIONS DEVELOPMENT IIIB
SUBJECT CODE/S :	APDB301
DATE :	17 SEPTEMBER 2025
DURATION :	1.5 HOURS
TIME :	10H00 - 11H30
TOTAL MARKS :	80
NUMBER OF PAGES :	8 (INCLUDING THE COVER PAGE)
EXAMINER/S :	MR. ISAIAH ADEBAYO

INSTRUCTIONS/REQUIREMENTS: -

[1] ANSWER ALL 40 MCQ QUESTIONS BY DETACHING THE LAST PAGE OF THE QUESTION PAPER AND COMPLETING YOUR DETAILS ON THAT PAGE. YOU ARE REQUIRED TO SUBMIT THE LAST PAGE AT THE END OF THE SESSION.

[2] USE THE MARK ALLOCATION AS GUIDE TO ESTIMATE THE TIME SPENT ON A QUESTION. EACH QUESTION HAS BEEN ASSIGNED EQUAL WEIGHT OF 2 MARKS. TOTAL MARKS (40 * 2)

Do not turn the page until permission is given

MCQ Total Marks **[80 marks]**

1.1 Which of the following best distinguishes synchronous from asynchronous programming?

a) In synchronous programming, tasks always run faster.

b) Asynchronous programming allows tasks to run concurrently without blocking the main thread.

c) Synchronous programming always requires multiple threads.

d) Asynchronous programming eliminates the need for error handling.

1.2 In a GUI-based application, why is asynchronous programming often preferred?

a) To increase CPU clock speed.

b) To prevent the UI from freezing during long-running operations.

c) To automatically optimize memory allocation.

d) To bypass compiler checks.

1.3 Which of the following correctly describes an async method in C#?

a) It runs on a separate thread by default.

b) It returns immediately with a placeholder, typically a Task or Task<T>.

c) It cannot contain await expressions.

d) It must always return void.

1.4 What happens inside an async method when the await keyword is encountered?

a) The current thread is paused, and control is returned to the caller until the awaited task completes.

b) The method blocks until the task is finished.

c) A new thread is always spawned to handle the continuation.

d) The compiler ignores the await keyword during execution.

1.5 Which of the following is a key **misconception** about asynchronous programming?

a) Async code improves responsiveness in UI applications.

b) Async programming guarantees faster execution.

c) Async code avoids blocking the main thread.

d) Async programming is compiler-assisted with state machines.

1.6 What is the primary advantage of using asynchronous programming in C#:

a) simplifies debugging

b) increase code complexity.

c) Improves application responsiveness.

d) Reduces memory usage.

Examine the following code and then answer questions 1.7-1.10

```
static async Task ProcessData()
{
    Console.WriteLine("Processing started");
    await Task.Delay(1000);
    Console.WriteLine("Processing completed");
}

static void Main(string[] args)
{
    var task = ProcessData();
    Console.WriteLine("Main completed");
    Console.Read();
}
```

1.7 What will be the last output shown in the console when this program is executed?

a) Processing started

b) Processing completed

c) Main completed

d) None of the above

1.8 Why does the message "Main completed" appear before "Processing completed" in this program?

a) Because await Task.Delay(1000) runs synchronously on the same thread.

b) Because await schedules the continuation of ProcessData after yielding control back to the caller.

c) Because ProcessData is executed entirely in a new thread that blocks the main thread.

d) Because Console.Read() forces "Processing completed" to be delayed until the user presses Enter.

1.9 When var task = ProcessData(); is executed in Main, what is the state of the returned Task immediately after the call?

a) TaskStatus.Created — it hasn't started until awaited.

b) TaskStatus.WaitingForActivation — because the task needs explicit Start().

c) TaskStatus.Running — it begins execution synchronously until the first await.

d) TaskStatus.Canceled — since the task has no continuation yet.

1.9.0 If Main ended without the Console.Read(), what would most likely happen?

a) "Processing completed" might not print, because the process could terminate before the awaited task finishes.

b) Both "Main completed" and "Processing completed" would always be printed.

c) The program would throw an exception because ProcessData() wasn't awaited.

d) "Processing completed" would be printed immediately before "Main completed".

2.1. Which of the following best describes Azure Service Bus?

a) A database service for structured data storage.

b) A load balancer for distributing HTTP requests.

c) A real-time streaming service for video delivery.

d) A fully managed messaging broker for reliable asynchronous communication.

2.2. What is the primary benefit of using Azure Service Bus for decoupling systems?

a) It ensures both sender and receiver must be online simultaneously.

b) It stores data permanently in SQL storage.

c) It allows the sender and receiver to work independently without time synchronization.

d) It reduces the need for authentication between services.

2.3. In Service Bus, which of the following patterns ensures **one consumer per message**?

a) Topics

b) Subscriptions

c) Queues

d) Event Hubs

2.4. A financial application publishes stock price updates. Multiple consumers (trading bots, dashboards, audit logs) each need a copy of every message. Which Service Bus entity is most appropriate?

a) Queue

b) Topic with multiple subscriptions

c) Single subscription with filtering

d) Dead-letter queue

2.5. Which analogy correctly describes the Service Bus namespace?

a) A single envelope containing all letters.

b) A post office building containing many mailboxes.

c) A letter carrier delivering mail.

d) A mailbox assigned to a single recipient.

2.6. Which of the following access control mechanisms can be defined at the namespace level?

a) Role-based access control (RBAC) and Shared Access Signatures (SAS).

b) SQL authentication using usernames and passwords.

c) OAuth only.

d) No authentication, since Service Bus is internal only.

2.7. An IoT system sends millions of telemetry messages to Service Bus. Multiple consumers (analytics engine, dashboards, alerts) must process them independently. Which pattern is best?

a) Queue

b) Topic with subscriptions

c) Single consumer queue

d) Database table insert

2.8. If a Service Bus queue is used for background task scheduling, what happens if multiple workers are polling the same queue?

a) Every worker will receive a copy of each message.

b) Each message is delivered to exactly one worker.

c) The queue automatically merges duplicate tasks.

d) Tasks are executed in parallel by default.

2.9. Which Service Bus capability allows you to send a message that should only become visible to receivers at a specific time in the future?

a) Delayed queues

b) Retry policy

c) Scheduled delivery

d) Message sessions

2.9.0. Which of the following is a **key difference** between queues and topics in Azure Service Bus?

a) Queues guarantee faster message delivery than topics.

b) Queues support only text messages, topics support binary.

c) Queues are point-to-point, topics support publish-subscribe with multiple subscribers.

d) Topics are stored in SQL, queues in blob storage.

2.9.1. A company wants to isolate messaging traffic between two teams working on separate applications. What is the best practice in Service Bus?

a) Create separate queues inside a single namespace.

b) Create separate topics inside a single namespace.

c) Use separate namespaces for each team.

d) Use filters on subscriptions to isolate traffic.

2.9.2. What is the purpose of Transparent Data Encryption in Azure SQL?

a) To encrypt data at rest

b) To encrypt data during transit

c) To provide two-factor authentication

d) To create database backups

2.9.3. Is the following statement TRUE or FALSE? "In a topic, multiple subscribers can receive a copy of the same message."

a) True

b) False

2.9.4. Which of the following BEST describes the role of a Queue in Azure Service Bus?

a) Enables broadcast messaging to multiple recipients

b) Facilitates direct communication without intermediaries

c) Ensures reliable one-to-one message delivery

d) Acts as a storage for messages that are never processed

2.9.5. When using Service Bus Queues, messages are automatically deleted from the queue once they are received by a consumer, even if the consumer hasn't processed them yet.

a) True

b) False

3.1. Which of the following best defines a Web API?

a) A desktop application for sending emails.

b) A database for storing client-side data.

c) A set of rules and endpoints that allow software systems to communicate over the web.

d) A protocol for real-time video streaming.

3.2. Why are Web APIs important for modern applications?

a) They allow different applications to communicate and share data efficiently.

b) They force frontend and backend to be tightly coupled.

c) They are only useful for mobile apps.

d) They eliminate the need for servers.

3.3. Which of the following formats is most commonly used for Web API responses?

a) CSV

b) JSON

c) DOCX

d) EXE

3.4. A mobile app needs real-time weather data from a remote server. Which role does the Web API play?

a) It stores the weather data locally on the device.

b) It fetches and delivers the weather data from the server to the app.

c) It replaces the need for the mobile app.

d) It directly updates the device's operating system.

3.5. Which of the following is NOT a core benefit of using Web APIs?

a) Data sharing across platforms

b) Separation of frontend and backend

c) Reusability of server logic

d) Automatic UI design generation

3.6. What does REST stand for?

a) Representational State Transfer

b) Remote Execution Standard Technology

c) Reliable Server Transfer

d) Reusable Endpoint Service Type

3.7. Which statement correctly describes RESTful services?

a) They are a protocol that defines how HTTP must be used.

b) They follow design principles for stateless, resource-based communication over HTTP.

c) They require storing session state on the server.

d) They can only return data in XML format.

3.8. Which HTTP method is used to update an existing resource in REST?

a) GET

b) POST

c) PUT

d) DELETE

3.9. Which REST principle ensures that every client request contains all the information needed and the server stores no session information?

a) Resource-based

b) Stateless

c) Uniform Interface

d) Representation of Resources

3.9.0 If the endpoint is GET /users/1, what is the expected action?

a) Delete the user with ID 1

b) Create a new user with ID 1

c) Retrieve the details of the user with ID 1

d) Update the user with ID 1

3.9.1. In RESTful APIs, what identifies a specific resource?

a) HTTP method

b) URI (Uniform Resource Identifier)

c) JSON format

d) SSL certificate

3.9.2. Which of the following is a key advantage of RESTful services?

a) They are heavy and complex but highly secure.

b) They can only be implemented in C# or Java.

c) They require proprietary client software.

d) They are lightweight, fast, and platform-independent.

3.9.3. Which HTTP method would you use to remove a resource in a RESTful API?

a) GET

b) POST

c) PUT

d) DELETE

3.9.4. RESTful services require the server to maintain session state for each client request to function correctly.

a) True

b) False

3.9.5. In REST, the HTTP method `POST` is typically used to create a new resource on the server.

a) True

b) False